

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	SWTID-2026-6128
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution, with purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	Dual-model emotion detection	Classifies free text into 5 emotions (Confused, Frustrated, Curious, Confident, Bored) with confidence scoring.	emotion_detection	Done	High
2	Mixed-emotion detection	Flags secondary emotions above a confidence threshold alongside the primary prediction.	emotion_detection	Done	Medium
3	Gemini-powered guidance generation	Produces personalized guidance text based on the detected emotion.	ai_response	In Progress (mock only)	High
4	Offline fallback handler	Returns canned guidance if the Gemini API is unavailable.	ai_response	Done	Medium
5	CSV interaction logging	Persists every interaction (input, emotion, confidence, response) for later analysis.	logging_system	Done (schema cleanup pending)	Medium
6	Analytics dashboard	Visualizes emotion distribution and BiLSTM vs. BERT confidence via Plotly.	analytics_dashboard	Done	Medium
7	Model comparison	Runs both BiLSTM and BERT on the same input and reports agreement.	emotion_detection	Done	Low
8	Streamlit single-page UI	Ties all modules together into one interactive input-to-guidance flow.	app.py	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7 (6 fully done, 1 in progress — Gemini integration)
Core / Must-Have Features	4 (Emotion Detection, Guidance Generation, Logging, Streamlit UI)
Additional / Nice-to-Have Features	4 (Mixed-Emotion Detection, Offline Fallback, Analytics Dashboard, Model Comparison)
Features Tested & Verified	3 (Emotion detection pipeline, CSV logging, Streamlit UI flow — Gemini integration not yet verified since it runs on the mock handler)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Streamlit input form, results display, and analytics dashboard
2	Backend / Logic	4	Emotion detection, mixed-emotion detection, model comparison, guidance generation
3	Database / Storage	1	CSV interaction logger
4	API / Integration	2	Gemini API client, offline fallback handler
5	Security / Authentication	0	Not yet implemented — no login is required for the current prototype